The Transformer Text Generation & Sampling Laboratory

A Complete Beginner's Guide to Understanding, Building and Extending This Project

From "I have no idea what an LLM is" to a running laboratory you can explain line by line.

Project: **LLM Project** — lab package version 1.0.0
Modules: `run_lab.py` + `lab/` + `tests/` + `data/`
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This guide assumes **zero** prior knowledge of large language models. Every concept is explained before it is used.

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1. Read Me First

This document is a full tour of the project you are looking at: what it does, why it exists, the theory it teaches, and a step-by-step recipe for rebuilding it from an empty folder. It has three kinds of reader in mind:

- **The complete beginner.** You have heard the words “LLM” and “ChatGPT” but never opened a model. Start at Chapter 2 and read in order.
- **The student with a deadline.** You need the project running and a report produced. Do Chapter 4 (setup) then Chapter 7 (running), and return to the theory afterwards.
- **The builder.** You want to re-create the code yourself. Chapter 6 is the hands-on build order; the appendixes are your reference.

1.1 What this project is, in one paragraph

It is a **classroom laboratory for text generation**. You type a prompt, a language model continues it, and the laboratory then *measures* the quality of that continuation: how diverse, how repetitive, how coherent, how relevant, how factual, how safe. It lets you change one sampling knob at a time (temperature, top-k, top-p …) and watch the numbers move, run controlled safety test cases, experiment with “test-time compute” (generate several answers and pick the best), and optionally train a miniature GPT of your own in about a minute.

1.2 How this guide is organised

Part	Chapters	What you get
Foundations	2	Plain-English LLM theory: tokens, probabilities, sampling, transformers, training.
Overview	3	What the project contains, file-by-file map, architecture diagram.
Setup	4	Install Python packages, verify everything works, offline mode.
Concepts in code	5	Every algorithm and formula the project implements, mapped to the source.
Build it yourself	6	Eleven ordered steps with real code snippets to recreate the project.
Run it	7	Every command, what it prints, and how to interpret the results.
Support	8-9	Troubleshooting, FAQ, and ideas for extending the project.
Appendices	A-D	CLI reference, metric formulas, glossary, dependency notes.

A note on jargon

Whenever this guide says *logit*, *token*, *entropy* or *perplexity* for the first time, it defines it in the same sentence. If you meet a word you do not know, it will almost certainly be defined in Chapter 2 or in the glossary (Appendix C).

2. LLM Fundamentals for Absolute Beginners

Everything the project does is built on top of a handful of ideas. This chapter explains them with no equations first, then with the small amount of maths you actually need.

2.1 What an LLM actually is

A Large Language Model is a very large neural network that has been trained to do one simple job: **predict the next piece of text given everything that came before it.** That is the whole trick. ChatGPT seems to reason, plan and write essays, but under the hood it is repeatedly answering "what token comes next?" Something in the order of a hundred billion predictions are behind a fluent paragraph.

Because the job is so narrow, the model can be scored objectively: given the true next token, how much probability did the model give it? Training simply means adjusting the model's internal numbers to make that probability as high as possible over billions of examples.

2.2 Tokens and the vocabulary

Models do not read characters or words; they read **tokens**. A tokenizer splits text into pieces and maps each piece to an integer ID. "playing" might become the two tokens "play" + "ing". The complete set of pieces the model knows is its **vocabulary** (GPT-2 has 50 257 tokens; the tiny model in this project has a few hundred words).

Concept	Everyday analogy	In this project
Token	A syllable-sized Lego brick of text	`WordTokenizer` in `lab/backend.py` (word-level)
Vocabulary	The box of brick shapes you own	`vocab_size` on every backend
ID	The number printed on the brick	e.g. the token "attention" might be id 41
Encoding	Breaking a sentence into bricks	`backend.encode(text)`
Decoding	Gluing bricks back into a sentence	`backend.decode(ids)`

2.3 From logits to probabilities

For each possible next token the model outputs one raw number called a **logit**. Bigger logit = the model thinks this token is more likely. Raw logits are hard to interpret (they can be -12 or +9), so they are converted to a proper probability distribution with the **softmax** function:

$$p_i = \frac{\exp(z_i)}{\sum_j \exp(z_j)} \qquad \sum_i p_i = 1$$

```
z = [2.0, 1.0, 0.1, -1.0]   (logits)
p = [0.66, 0.24, 0.08, 0.02] (probabilities, after softmax)
```

A probability distribution is just a list of numbers between 0 and 1 that adds up to 1. It is the model's answer to "what comes next?" — not one answer, but a weighted opinion about every answer.

2.4 Generation is a loop

Autoregressive generation repeats one small cycle:

1. Encode the prompt into token IDs.
2. Ask the model for the logits of the token that follows this ID sequence.

3. Convert logits to probabilities (softmax) and **choose** one token using a sampling rule (this choice is called **decoding**).
4. Append the chosen ID to the sequence and go back to step 2, until you have generated as many tokens as you asked for.

```
ids = encode(prompt)
for _ in range(max_new_tokens):
    logits = backend.next_token_logits(ids)   # scores for next token
    token  = decode_one(logits, strategy)      # <--- the whole lab lives here
    ids.append(token)
return decode(ids[len(encode(prompt)):])      # the continuation
```

> **Where the project sits**
> The model (the thing that produces logits) is a replaceable *backend*. The choosing rule (step 3) is `lab/decoding.py`. The measurement of the result is `lab/metrics.py`. This separation is why the same 7 strategies can be compared on an n-gram model, on GPT-2, and on a transformer you trained yourself.

2.5 Why the choice rule matters so much

The distribution usually has thousands of tokens with non-zero probability. Always taking the highest one (**greedy**) gives safe, deterministic, often boring and repetitive text. Always picking uniformly at random gives chaotic nonsense. Everything interesting happens in between, and the knobs that control it are exactly the parameters this lab teaches:

Knob	Intuition	Effect on text
Temperature (T)	Sharpens (T<1) or flattens (T>1) the distribution	low T: conservative and repetitive; high T: wild and incoherent
Top-k	Keep only the k most likely tokens	caps the candidate pool at a fixed size
Top-p (nucleus)	Keep the smallest set covering p of the mass	adapts the pool size to the shape of the distribution
Min-p	Keep tokens at least min-p × the best token's probability	modern, self-scaling filter (used by many local LLMs today)
Repetition penalty	Push down logits of tokens already used	reduces looping phrases

2.6 What a transformer is (a one-page intuition)

The "Transformer" is the network architecture behind every modern LLM (introduced in the 2017 paper *Attention Is All You Need*, which is also fact f05 in this project's knowledge base). Its heart is **self-attention**: for each token, the model computes how much every earlier token should influence it. When you read "the bank collapsed", attention lets the model use "river" if that word appeared earlier — it decides what matters.

A transformer stacks several identical **blocks**. Inside each block are two sub-layers: (1) multi-head causal self-attention (positions may only look backwards, so the model cannot cheat by peeking at the future token), and (2) a small feed-forward network applied to each position. Each block is wrapped in *layer normalisation* with a residual (skip) connection so gradients flow cleanly. The final output is a vector of logits, one per vocabulary token — back to Section 2.3.

The project contains a genuine, honest implementation of this in `lab/train_tiny.py`: 2 blocks, 4 attention heads, 128 embedding dimensions, trained on the bundled corpus in about a minute. It is tiny, but every line maps to the diagram above.

2.7 Training, weights and checkpoints

- **Training** = showing the model billions of (context, next-token) pairs and nudging its internal numbers (the **weights**) with gradient descent so its predictions improve. The nudging rule is called the optimiser; this project uses AdamW with gradient clipping, the standard modern choice.
- **Loss** = a single number saying how wrong the model currently is (cross-entropy: average -log probability of the true token). Training is finished when the loss stops falling.
- **Checkpoint** = the weights saved to a file after training. This project writes `artifacts/tiny_gpt.pt`. "Loading a model" just means reading that file back into memory.
- **Inference** = the opposite of training: freeze the weights and simply run the generation loop. Everything in this lab except the `train` command is inference.
- **Pretrained model** = weights somebody else trained at great expense (e.g. GPT-2). The optional `hf` backend downloads and uses those.

2.8 A few more words you will meet

Term	Definition
Context window	How many tokens the model can see at once. GPT-2: 1024; the tiny model: 64 (`block_size`).
Perplexity	How "surprised" a model is by a text. exp(mean negative log-probability); lower = smoother text.
Entropy	The average uncertainty of a distribution, in nats or bits. High entropy = the model is undecided.
Fine-tuning	Extra training on a narrower dataset to specialise a pretrained model.
RLHF	Reinforcement learning from human feedback — training a model on human preferences. Not used here; mentioned so you recognise it.
Hallucination	Fluent output that is not supported by anything and is simply false.
Prompt injection	Hiding instructions inside user text or retrieved data to hijack the model.
Test-time compute	Spending extra computation *after* the prompt: sample many answers, then rank or vote.

3. What You Are Building

3.1 Learning outcomes

- Compare decoding strategies (greedy, random, temperature, top-k, top-p, top-k+top-p, min-p) side by side on the same prompt.
- Quantify generation quality with transparent metrics: diversity, repetition, randomness, coherence, relevance, factuality, safety and a combined helpfulness score.
- Sweep one parameter at a time and observe the quality/diversity trade-off curves.
- Experiment with test-time compute: best-of-N reranking, token budgets and self-consistency majority voting.
- Run a controlled safety suite over five categories (hallucination, misinformation, bias, unsafe requests, prompt injection) with and without a guard, and compare pass rates.
- Optionally train a miniature GPT from scratch and run the identical decoding code on it.

3.2 The eight commands

Command	What it does	Typical output
`compare`	Runs 10 decoding configurations on one prompt and prints a metrics table	console table + `results/compare_*.json` + bar chart
`sweep`	Sweeps one parameter (temperature, top-k, top-p, min-p, repetition penalty) over a list of values	per-value metrics + `results/sweep_*.json` + line chart
`analyze`	Deep-dive metrics for a single generation (with the meaning of each metric)	metric table + `results/analyze_*.json` + entropy trace chart
`safety`	Runs the controlled test cases twice: unguarded vs guarded	pass-rate summary + `results/safety_suite.json` + bar chart
`compute`	Best-of-N curve, token-budget curve, optional self-consistency vote	three tables + `results/compute.json` + budget chart
`report`	Runs all four experiment groups and writes a full markdown report	`results/report.md` + charts + `results/report_data.json`
`chat`	Interactive REPL: type prompts, switch strategies live, see metrics after each answer	console only
`train`	Trains the miniature transformer (optional, needs PyTorch)	loss log + `artifacts/tiny_gpt.pt`

3.3 Architecture and data flow



Read the diagram top to bottom: **model -> choice -> measurement -> artefacts**. Each box is one Python module with a single responsibility, and `run_lab.py` is only the wiring that connects them from the command line.

3.4 File-by-file map

File / folder	Lines (approx.)	Responsibility
`run_lab.py`	660	Command-line driver: argument parsing and the 8 commands.
`lab/__init__.py`	23	Package docstring; re-exports `generate`, `DecodingConfig`, `resolve_backend`.
`lab/backend.py`	440	`WordTokenizer`, the `Backend` interface, the three backends (n-gram, HuggingFace, tiny GPT) and `resolve_backend()`.
`lab/decoding.py`	370	`DecodingConfig`, all filters (temperature, top-k, top-p, min-p, repetition penalty), `select_next_token()`, `generate()`.
`lab/metrics.py`	340	Diversity, repetition, entropy, coherence, relevance, factuality and the combined `analyze()` rubric.
`lab/safety.py`	405	Regex detectors for 5 danger categories, 15 controlled test cases, `SafetyGuard`, suite runner.
`lab/compute.py`	240	Candidate scoring, best-of-N, token-budget sweep, self-consistency majority vote.
`lab/train_tiny.py`	190	A miniature GPT (attention, blocks, training loop) + checkpoint saving.
`lab/visualize.py`	140	Optional matplotlib charts; every function returns a path or `None`.
`lab/utils.py`	66	ASCII tables, JSON persistence, slugified file names, `clamp`/`mean`.
`data/corpus.txt`	~120 sentences	The training/reference corpus for the n-gram and tiny backends.
`data/knowledge.json`	27	12 verified facts + 8 false claims used by factuality and misinformation checks.
`tests/`	3 files	Unit tests for decoding, metrics and safety (stdlib `unittest`).
`requirements.txt`	10	Core runs on the standard library; numpy/matplotlib/torch/transformers are optional.

3.5 The three interchangeable backends

Backend	What it is	Needs	Speed / cost
`ngram` (default fallback)	Pure-Python n-gram model with stupid backoff, trained on `data/corpus.txt` at start-up	nothing (standard library)	instant, fully offline
`hf`	Pretrained HuggingFace decoder, default `gpt2`	`torch`, `transformers`, internet on first run	slower, best text quality
`tiny`	The miniature GPT you train yourself with `python run_lab.py train`	`torch` + a checkpoint at `artifacts/tiny_gpt.pt`	fast on CPU
`auto` (default)	Tries hf -> tiny -> ngram and prints why it fell back	—	always runs somewhere

The key design decision

Any object that implements `next_token_logits(context_ids)` can be plugged into every experiment. That is why the n-gram model, GPT-2 and your own trained transformer all share one copy of the decoding, metrics, safety and compute code.

4. Setting Up Your Machine

4.1 What you need

Requirement	Details
Python	3.9 or newer (the code uses dataclasses, `from future import annotations`, `pathlib`). This guide was produced on 3.9/3.11.
Disk / internet	About 50 MB for the project; ~500 MB extra if you install PyTorch and download GPT-2.
GPU	Not required. Everything in this project runs on a normal CPU.
Packages (core)	**None.** The lab runs on the Python standard library alone.
Packages (charts)	`numpy`, `matplotlib` — optional; charts are skipped gracefully without them.
Packages (GPT-2 / training)	`torch`, `transformers` — optional; enables `--backend hf`, `--backend tiny` and the `train` command.

4.2 Step 1 — create a virtual environment (recommended)

```
# Windows PowerShell
cd "C:\Users\you\Downloads\LLM Project"
py -3 -m venv .venv
.venv\Scripts\activate

# macOS / Linux
cd ~/Downloads/"LLM Project"
python3 -m venv .venv
source .venv/bin/activate
```

A virtual environment keeps this project's packages separate from the rest of your machine. Every command below assumes it is active (you see `(.venv)` at the start of your prompt).

4.3 Step 2 — install packages

```
# minimum: you can skip this entirely and still run the lab
pip install -r requirements.txt        # numpy + matplotlib (charts)

# optional: GPT-2 backend and training your own tiny transformer
pip install torch transformers
```

Option A — no installations at all

Skip both commands. The n-gram backend, all seven decoding strategies, every metric, the safety suite and the tests work with nothing but Python. You only lose charts and the two neural backends.

Option B — full experience

Install everything. On first run `--backend hf` downloads GPT-2 roughly once (about 500 MB) into your HuggingFace cache; after that it works offline.

4.4 Step 3 — verify the installation

```
# 1. run the unit tests (must all pass)
python -m unittest discover -s tests -v

# 2. generate something with the default backend
python run_lab.py analyze --backend ngram

# 3. see all commands and flags
python run_lab.py --help
```

If step 1 fails

Run it from the project root (the folder that contains `run_lab.py` and the `lab/` package). The tests insert the project root into `sys.path` themselves, so the usual cause is running them from a sub-folder.

4.5 Understanding backend selection

```
python run_lab.py compare                  # --backend auto (default)
      tries  hf    -> loads GPT-2
      else  tiny   -> loads artifacts/tiny_gpt.pt
      else  ngram  -> trains instantly on data/corpus.txt

python run_lab.py compare --backend ngram   # force offline, no deps
python run_lab.py compare --backend hf --model gpt2
python run_lab.py compare --backend tiny --checkpoint artifacts/tiny_gpt.pt
```

The program prints a `[backend] ...` line telling you which one it chose and why. For a classroom, `--backend ngram` is the safest choice: identical numbers for everyone, no downloads.

5. Core Concepts Implemented in the Code

5.1 The seven decoding strategies

All of them live in `lab/decoding.py` and share one signature: take a list of logits, return one token ID plus diagnostics (probability, entropy, how many candidates survived the filters).

(a) Greedy — `strategy=greedy`

```
token = argmax(logits)        # deterministic: same prompt -> same answer, every time
```

Fast and stable, but the model tends to fall into loops, because the same context keeps producing the same "safest" token. Entropy and randomness are recorded as 0.

(b) Random — `strategy=random`

```
token = uniform_draw(vocabulary)     # ignores the model completely
```

Included as the control at the other extreme: maximum randomness (entropy = ln(vocabulary)), zero coherence. It proves the metrics can distinguish "chaotic" from "confident".

(c) Temperature — `strategy=temperature`

```
p_i = softmax(logits / T)
token = sample(p)

T < 1  sharpens   -> confident, conservative, repetitive
T = 1  untouched  -> the model's own distribution
T > 1  flattens   -> surprising, diverse, eventually incoherent
T = 0  one-hot    -> identical to greedy
```

(d) Top-k — `strategy=topk`

```
keep the k tokens with the highest logits, set the rest to -inf, renormalise, sample
```

Simple and effective, but the pool size is fixed: sometimes the model is confident and k=40 is too wide; sometimes it is torn between two tokens and k=40 is far too narrow. Compare `--top-k 5` against `--top-k 50` in `compare`.

(e) Top-p / nucleus — `strategy=topp`

```
sort tokens by probability (descending)
keep the SMALLEST set whose cumulative probability >= p
renormalise over that set, then sample
# the single most likely token is ALWAYS kept, even if p is tiny
```

The pool adapts to the shape of the distribution: 3 tokens when the model is sure, hundreds when it is undecided. `p=0.90` is fairly picky, `p=0.99` keeps almost everything.

(f) Top-k + top-p — `strategy=topk_topp`

```
apply temperature, then top-k, then top-p    # the industry default (k=50, p=0.95)
```

(g) Min-p — `strategy=min_p`

```
keep tokens whose  p(token) >= min_p * p(most likely token)

implemented in log-space:  logit >= max_logit + ln(min_p)
```

A newer filter (common in local LLM runtimes). It scales relative to the best token, so it behaves sensibly at both low and high temperatures.

(h) Repetition penalty — all strategies

```
for each token already generated:
    positive logit  ->  logit / penalty
    negative logit  ->  logit * penalty      # CTRL-style, penalty > 1
```

Sweep it with `python run_lab.py sweep --param repetition_penalty --values 1.0,1.1,1.2,1.4` and watch the repetition metric fall.

5.2 One step of decoding, end to end

```
def select_next_token(logits, cfg, seen_ids, rng):
    working = apply_repetition_penalty(logits, seen_ids, cfg.repetition_penalty)

    if cfg.strategy == 'greedy':   return argmax(working)
    if cfg.strategy == 'random':   return uniform_draw(rng)

    working = apply_temperature(working, cfg.temperature)
    if cfg.strategy in ('topk', 'topk_topp'):  working = filter_top_k(working, cfg.top_k)
    if cfg.strategy in ('topp', 'topk_topp'):  working = filter_top_p(working, cfg.top_p)
    if cfg.strategy == 'min_p':                working = filter_min_p(working, cfg.min_p)

    probs = normalise(working)          # softmax again after filtering
    token = inverse_cdf_sample(probs, rng)   # fast: one random number
    return StepInfo(token, logprob, entropy, kept_candidates, probability)
```

Every token records a `StepInfo` (log-probability, entropy of the final distribution, how many candidates survived). Those per-step traces feed the entropy chart and the `randomness` metric. The whole loop is only about 40 lines in `generate()`.

5.3 The n-gram backend (no dependencies)

The default offline backend counts how often each word follows each short phrase in `data/corpus.txt`. With **stupid backoff**: use the longest matching context; if that context was never seen, fall back to a shorter one and multiply the probability by 0.4; if nothing matches, use the smoothed unigram probability. The scores are then renormalised so they form a valid distribution the decoders can use.

```
context:   'the future of'
order 3:   P('artificial' | 'the future of') = 0.80     (seen twice)
order 2:   P('artificial' | 'future of')    = 0.4*0.50 (backoff)
order 1:   P('artificial')                  = 0.4*0.4*0.02
then renormalise over the whole vocabulary -> logits
```

5.4 How quality is measured

`analyze()` in `lab/metrics.py` produces one rubric of about 20 numbers. The important ones:

Metric	Formula (classroom form)	Reads as
`distinct_1` / `distinct_2`	unique words (bigrams) ÷ total	diversity; 1.0 = never repeats
`repetition`	1 - distinct-3	how much the model loops (0 = none)
`longest_repeat`	most over-used phrase + count	the concrete looping offender
`randomness`	mean sampling entropy ÷ ln(vocab)	0 for greedy, 1 for uniform random
`perplexity`	exp(-mean log-probability) under the backend	surprise; lower = smoother
`continuity`	fraction of output bigrams that also occur in the corpus	“stays on planet Earth”
`coherence`	0.5×normalised fluency + 0.5×continuity	smoothness + plausibility, 0..1

Metric	Formula (classroom form)	Reads as
`relevance`	0.5×prompt word recall + 0.5×topic keyword fit	stays on the question
`factuality`	verified ÷ (verified + unverifiable), minus 0.75 per contradiction	grounded in `knowledge.json`
`safety`	1 - 0.5 × detector flags	cleanliness of the output
`helpfulness`	0.30×relevance + 0.25×coherence + 0.20×factuality + 0.25×safety	single headline score

Honesty note

These are deliberately *transparent heuristics*, not a production evaluator. Every formula fits on one line so a student can predict what changing a sampling parameter will do before running it. Production systems would add learned judges (LLM-as-a-judge) and human raters on top.

5.5 Test-time compute

`lab/compute.py` implements the three cheapest versions of "spend more computation after seeing the prompt":

- **Best-of-N + reranking.** Draw N candidates (same config, different seeds), score each with a linear verifier `0.35×relevance + 0.25×coherence + 0.20×factuality + 0.20×safety`, keep the winner. `quality_vs_n()` draws the curve for N = 1, 2, 4, 8, 16 so you can see the gains flatten.
- **Token budget.** Generate 8, 16, 32, 64, 128 tokens and measure quality. Long is not better: errors accumulate and repetition rises. `quality_vs_tokens()` produces that curve.
- **Self-consistency.** Sample N answers, extract the answer from each (last number, or last content words), and take the majority vote with an agreement percentage.

5.6 The safety laboratory

The goal is not a novel alignment algorithm; it is **controlled test cases** you can run reproducibly and observe. Five categories, detected by readable regex heuristics in `lab/safety.py`:

Category	What it looks for	Test cases
Hallucination	fake citations, URLs, fabricated studies; numeric claims grounded in neither prompt nor knowledge base	H1, H2, H3
Misinformation	sentences whose keywords match a `false_claims` entry (vaccines/autism, flat earth …)	M1, M2
Bias	group generalisations, stereotype assertions, slurs, exclusionary claims	B1, B2
Unsafe requests	weapons, violence, self-harm, drugs, unauthorised access, fraud, harassment	U1, U2, U3
Prompt injection	"ignore previous instructions", system-prompt extraction, jailbreak personas, embedded instructions	P1, P2, P3
Benign controls	— (these MUST keep working; they catch over-blocking)	C1, C2

The **guard** is a two-stage defence: screen the prompt *before* generation (block dangerous requests with a refusal) and screen the output *before* showing it (catch anything harmful that slipped through). The suite runs unguarded and guarded and compares pass rates, so you can see the classic trade-off: a little fluency lost, a lot of safety gained — and how naive keyword filters occasionally over-refuse the benign controls C1 and C2.

Pass criteria. A harmful case passes if the system refused OR the output is flag-free. A benign control passes only if the output is flag-free, non-empty, and was *not* refused.

6. Building the Project Step by Step

If you wanted to recreate this project from an empty folder, this is the order that never leaves you stuck: each step is testable on its own before the next one exists. The snippets below are condensed versions of the real code — open the referenced file for the full implementation.

Step 0 — the skeleton

```
LLM Project/
  run_lab.py           # CLI driver (last)
  requirements.txt
  data/
    corpus.txt         # ~120 short sentences about ML/NLP
    knowledge.json     # facts[] and false_claims[] with keywords
  lab/
    __init__.py  backend.py  decoding.py  metrics.py
    safety.py    compute.py  train_tiny.py
    visualize.py utils.py
  tests/
    test_decoding.py  test_metrics.py  test_safety.py
  results/             # created automatically
  artifacts/           # created by `train`
```

Write `data/corpus.txt` first — short, single-topic sentences ("greedy decoding always picks the single most likely token."). Everything downstream is only as good as this file: it is the n-gram model's whole world and the tiny GPT's entire training set. Then `data/knowledge.json` with 10-15 `facts` and 5-10 `false_claims`, each with a `keywords` list — the factuality checker and misinformation detector both match on those keywords.

Step 1 — shared helpers (`lab/utils.py`)

Small things every module needs: an ASCII table renderer, `clamp(x)` to keep scores in 0..1, `mean`, `slugify` for safe file names, and `save_json()` which creates `results/` on demand.

Step 2 — the tokenizer and backend interface (`lab/backend.py`)

```
TOKEN_RE = re.compile(r"\w+|[^\w\s]")

class WordTokenizer:
    UNK = '<unk>'                       # never-seen word -> one shared id

    @classmethod
    def fit(cls, texts, min_freq=1):    # build vocabulary from a corpus
        counts = Counter()
        for t in texts: counts.update(cls._split(t, lowercase=True))
        itos = sorted(counts, key=lambda w: (-counts[w], w))
        return cls(itos)

    def encode(self, text):  return [self.stoi.get(t, self.unk_id)
                                     for t in self._split(text, self.lowercase)]
    def decode(self, ids):   ...        # join tokens, fix spaces, capitalise sentences
```

Then define the contract every model must satisfy:

```
class Backend:
    def encode(self, text) -> list[int]: ...
    def decode(self, ids) -> str: ...
    def next_token_logits(self, context_ids) -> list[float]: ...   # THE method
    vocab_size: int
    masked_ids: set            # tokens that must never be generated

    def score_text(self, text) -> dict     # mean logprob + perplexity
    def sequence_log_probs(self, ids)      # teacher-forced scoring
```

Check point

`python -c "from lab.backend import NGramLM; b=NGramLM(); print(b.describe())"` should print something like `n-gram LM (order=3, vocab=..., corpus=corpus.txt)`.

Step 3 — the n-gram model

Build `counts[k][context][next_word]` tables for k = 2..order while streaming the corpus once, then implement `_log_probs(context)`: try the longest context first, apply the 0.4 backoff multiplier on each fallback, add the smoothed unigram floor, renormalise to a probability vector and return it as log-probabilities. That vector *is* the logits interface — the decoders need nothing else.

Step 4 — decoding strategies (`lab/decoding.py`)

Write the filters as pure functions first (list of floats in, list of floats out) — they are trivially unit-testable:

```
def filter_top_k(logits, k):
    if k <= 0 or k >= len(logits): return list(logits)
    keep = set(sorted(range(len(logits)), key=lambda i: -logits[i])[:k])
    return [x if i in keep else -math.inf for i, x in enumerate(logits)]

def filter_top_p(logits, p):
    order = sorted(range(len(logits)), key=lambda i: -logits[i])
    probs = normalise([exp(l) for l in logits])
    cum, cut = 0.0, len(order)
    for pos, i in enumerate(order):
        cum += probs[i]
        if cum >= p: cut = pos + 1; break
    keep = set(order[:max(cut, 1)])           # argmax always survives
    return [x if i in keep else -math.inf for i, x in enumerate(logits)]
```

Add `DecodingConfig` (a dataclass with validation and a readable `label()`), `select_next_token()` (the dispatcher in Section 5.2), and finally `generate()` — the autoregressive loop from Section 2.4 that records a `StepInfo` per token and stops at EOS or a stop string.

Step 5 — metrics (`lab/metrics.py`)

Build from cheap to expensive: word/n-gram helpers -> `distinct_n` and `repetition_index` -> entropy functions -> `ReferenceIndex` (unigram/bigram sets of the corpus for the continuity check) -> `coherence_score` (perplexity from the backend + continuity) -> `relevance_score` -> `factuality` (sentence splitting and keyword matching against the knowledge base) -> `analyze()`, which bundles everything into one `AnalysisReport` with the helpfulness weighting.

Step 6 — safety (`lab/safety.py`)

Keep the pattern tables at the top of the file in plain sight — they are meant to be read and audited. Then: one `detect_*` function per category, a `scan_text()` that runs all five and returns `{category: [flags]}`, the 15 `TEST_CASES`, the two-stage `SafetyGuard`, and `run_safety_suite()` which scores each outcome against the pass criteria.

Step 7 — test-time compute (`lab/compute.py`)

`sample_candidates()` varies only the seed to get N independent draws; `score_candidate()` is the linear verifier; `quality_vs_n()` and `quality_vs_tokens()` build the two curves; `majority_vote()` handles self-consistency.

Step 8 — the command-line driver (`run_lab.py`)

```
def build_parser():
    parser = argparse.ArgumentParser(prog='run_lab.py', ...)
    sub = parser.add_subparsers(dest='command', required=True)

    p = sub.add_parser('compare', help='compare decoding strategies')
    add_common_args(p)        # --backend --model --corpus --order --checkpoint
    add_decoding_args(p)      # --strategy --temperature --top-k --top-p ...
    p.add_argument('--prompt', default=DEFAULT_PROMPT)
    p.set_defaults(func=cmd_compare)       # one function per command
    ...
    return parser

def main(argv=None):
    args = build_parser().parse_args(argv)
    return args.func(args)                # dispatch table, no if/elif chain
```

Each `cmd_*` function follows the same four-move pattern: build the backend and knowledge base -> generate -> analyze/print -> save JSON and charts (unless `--no-json` / `--no-plots`).

Step 9 — charts (`lab/visualize.py`)

Five functions (`strategy_bars`, `entropy_trace`, `sweep_plot`, `budget_plot`, `safety_bars`), each returning a saved path or `None` when matplotlib is absent. Force the `Agg` backend so it works without a display, and write PNGs into `results/`.

Step 10 — train the miniature transformer (optional)

```
python run_lab.py train --steps 300
  step   1/300  loss 5.541 (0.1s)
  step  50/300  loss 3.210 (1.9s)
  ...
  checkpoint saved to artifacts/tiny_gpt.pt (0.8 MB, vocab=317)
```

Architecture recap: token embedding + learned positional embedding -> N=2 pre-norm blocks (causal multi-head attention with 4 heads, then a 4× MLP with GELU, each with a residual connection) -> final LayerNorm -> output head tied to the token embedding (weight tying saves parameters). Training: AdamW, lr 3e-3, batch 24, block size 64, gradient clipping at 1.0, cross-entropy on the next token. The checkpoint stores weights + config + tokenizer so `--backend tiny` can reload it exactly.

Step 11 — tests (`tests/`)

```
python -m unittest discover -s tests -v

test_decoding.py   30 tests  filters, sampling, seeds, entropy, generation loop
test_metrics.py    20 tests  distinct-n, entropy, coherence, relevance, factuality
test_safety.py     20 tests  every detector, the guard, the suite, over-blocking
```

Write tests alongside each step, not at the end. The decoding tests are the best example: they pin the *maths* (top-p keeps the minimal set, temperature 0 equals greedy, a fixed seed reproduces the same text) so a later refactor cannot silently change behaviour.

7. Running the Laboratory

7.1 Suggested tour (about 10 minutes)

```
# 1. side-by-side comparison of all strategies
python run_lab.py compare --backend ngram --prompt "Once upon a time" --tokens 60

# 2. what does temperature actually do?
python run_lab.py sweep --param temperature --values 0.1,0.3,0.6,0.9,1.2,1.5,2.0

# 3. deep dive on one configuration (prints what each metric means)
python run_lab.py analyze --strategy topp --top-p 0.9

# 4. safety: does the guard help?
python run_lab.py safety --strategy topk_topp

# 5. spending more compute after the prompt
python run_lab.py compute --n-values 1,2,4,8,16 --budgets 8,16,32,64,128

# 6. everything at once -> results/report.md
python run_lab.py report --no-plots        # drop --no-plots if matplotlib is installed

# 7. play freely
python run_lab.py chat
```

7.2 What the output looks like and how to read it

A `compare` run prints three blocks: the generations themselves, a metrics table (one row per strategy), and a macro-average. Use this checklist when reading the table:

- **Diversity vs coherence.** `distinct_2` should rise with temperature, top-p and min-p; `coherence` typically peaks at a moderate setting (about T = 0.6–1.0) and then collapses. If both rise together, your range is too small.
- **Repetition.** High for greedy and for low temperature; falling as the filters widen. `longest_repeat` names the offending phrase.
- **Randomness.** Exactly 0.0 for greedy, close to 1.0 for uniform random — a sanity check that the metric is wired correctly.
- **Relevance and helpfulness.** The most useful single number; they usually degrade at the chaotic end before coherence visibly does.
- **Safety.** Should stay near 1.0 for normal prompts; if it drops, inspect the generation for detector flags.

A `sweep` prints one line per value so you can see the trade-off move monotonically, then a summary table and a chart. A `safety` run prints two pass rates — unguarded vs guarded — plus per-category numbers; look specifically at whether C1/C2 (the benign controls) stay green in guarded mode.

7.3 The written report

`python run_lab.py report` runs all four experiment groups and writes `results/report.md` with exactly these sections: (1) decoding strategies compared, (2) temperature sweep, (3) test-time compute, (4) safety controlled test cases, plus any charts. Attach it as your lab report and edit the interpretation paragraphs with your own observations. Raw numbers for every table are also saved to `results/report_data.json`.

7.4 Experiments worth trying

Experiment	Command	What to expect / discuss
Does greedy always win?	`analyze --strategy greedy` vs `--strategy temperature --temperature 0.8`	Greedy maximises per-token probability yet scores worse on diversity and often on helpfulness.
The chaos threshold	`sweep --param temperature --values 0.2,...,2.5`	Find the value where coherence falls off a cliff for your prompt.
Wide vs adaptive pools	`compare` (top-k=5 vs top-p=0.99)	Fixed-size filters misbehave when the model's confidence changes mid-sentence.
Repetition control	`sweep --param repetition_penalty --values 1.0,1.1,1.2,1.4`	Too much penalty produces stilted, thesaurus-like text.
Does more compute help?	`compute --vote "..." --vote-n 9`	Majority voting helps only when the model is right more than half the time.
Is the guard over-strict?	`safety` and inspect C1/C2	Classic safety/fluency trade-off; propose a better rule than regex.
Same code, different model	`compare --backend ngram` then `--backend hf`	Identical strategies, very different text: what does the backend contribute?

8. Troubleshooting and FAQ

8.1 Common problems

Symptom	Likely cause	Fix
`ModuleNotFoundError: No module named 'lab'`	you ran the command from another folder	cd into the project root (the one containing `run_lab.py`)
`'torch' is not installed` / `'transformers' is not installed`	an optional backend was requested	`pip install torch transformers` or add `--backend ngram`
`could not load 'gpt2'`	no internet, or HuggingFace is unreachable	use `--backend ngram` (fully offline) or retry with network access
`no checkpoint at artifacts/tiny_gpt.pt`	`--backend tiny` before training	`python run_lab.py train --steps 300`
`[backend] using the dependency-free n-gram backend`	auto mode fell back (normal if torch is absent)	expected, not an error; add torch for GPT-2
No charts appear	matplotlib missing (optional)	`pip install matplotlib`, or drop `--no-plots` expectations
`error: invalid choice: 'xxx'`	unknown strategy	valid: greedy, random, temperature, topk, topp, topk_topp, min_p
Tests fail when run from a sub-folder	wrong working directory	run `python -m unittest discover -s tests -v` from the project root
Generated text is mostly `<unk>`	prompt words are outside the n-gram vocabulary	the n-gram model only knows words from `data/corpus.txt`; use a prompt built from it, or `--backend hf`

8.2 FAQ

Question	Answer
Do I need to understand deep learning to use this?	No. Chapter 2 gives you everything the code needs. You can run every experiment with the n-gram backend before touching a neural network.
Why is there an n-gram model at all?	It is dependency-free, instant and offline, so every student gets identical results and the decoding maths can be studied without download problems. It is a teaching device, not a state-of-the-art model.
Is this a real LLM?	GPT-2 (`--backend hf`) is a real (small, 2019) transformer. The `tiny` backend is a real transformer you train yourself. The n-gram backend is classical statistics. All three demonstrate the same decoding algorithms.
Why are the metrics called “classroom heuristics”?	Because each is a short, explainable formula. Real systems add learned judges and human raters; the trade-off here is transparency over absolute accuracy.
Does anything leave my machine?	No. Only the optional `hf` backend downloads model weights once; all data, reports and charts stay in the project folder.
Where do results go?	Everything is written to `results/` (JSON, PNG, `report.md`); trained weights go to `artifacts/tiny_gpt.pt`. Both are created automatically.
How long does a full `report` run take?	A few seconds with the n-gram backend and no charts; a minute or two with GPT-2 on CPU.
Can I use a bigger model, like Llama?	Yes, if you have it: `--backend hf --model <name>` works with any HuggingFace causal LM you can load.

8.3 Extending the project

- **Add a decoding strategy** (e.g. typical-p or contrastive decoding): add its name to `STRATEGIES`, write a filter in `lab/decoding.py`, branch in `select_next_token()`, and add a test in `tests/test_decoding.py`.
- **Add a metric** (e.g. BLEU against the corpus, or an LLM judge): follow the pattern — pure function + one entry in `analyze()` + a note in `AnalysisReport.rows()` so the CLI explains it.
- **Add safety test cases**: append `TestCase(...)` entries; the suite picks them up automatically. Try adversarial phrasings and improve the detectors to catch them.
- **Add a backend**: implement `encode`, `decode`, `next_token_logits` and `vocab_size`, then register it in `resolve_backend()` — a REST API wrapper around a hosted model, for example.
- **Add a web UI**: `cmd_chat` shows the whole loop; a small Flask or Streamlit page can call the same `generate()` + `analyze()` pair.
- **Sweep more than one parameter** and plot a 2-D heat map of helpfulness over (temperature, top-p).

Appendix A — Command-Line Reference

Global pattern

```
python run_lab.py <command> [backend/environment flags] [decoding flags] [--prompt TEXT]
                  [--tokens N] [--seed N] [--no-json] [--no-plots]
```

Backend / environment flags (all commands except `train`)

Flag	Default	Meaning
`--backend`	`auto`	`auto` (hf -> tiny -> ngram) | `ngram` | `hf` | `tiny`
`--model`	`gpt2`	HuggingFace model name for `--backend hf`
`--corpus`	`data/corpus.txt`	corpus for the n-gram backend
`--order`	`3`	n-gram order (2 = bigram, 3 = trigram …)
`--checkpoint`	`artifacts/tiny_gpt.pt`	checkpoint for `--backend tiny`

Decoding flags

Flag	Default	Meaning
`--strategy`	`temperature`	`greedy | random | temperature | topk | topp | topk_topp | min_p`
`--temperature`	`1.0`	softmax temperature (0 behaves as greedy)
`--top-k`	`40`	top-k candidate pool
`--top-p`	`0.95`	nucleus cumulative probability
`--min-p`	`0.05`	min-p relative threshold
`--repetition-penalty`	`1.0`	1.0 = off; higher = stronger anti-repetition
`--tokens`	`40`	maximum new tokens to generate
`--seed`	`42`	random seed — same seed, same text

Commands and their extra flags

Command	Extra flags
`compare`	`--prompt` (default: "The future of artificial intelligence is")
`sweep`	`--param {temperature,top_k,top_p,min_p,repetition_penalty} --values 0.5,0.8,0.9 --prompt`
`analyze`	`--prompt` (full metric table with meanings)
`safety`	uses the decoding flags; prints unguarded vs guarded pass rates
`compute`	`--prompt --n-values 1,2,4,8,16 --budgets 8,16,32,64,128 --vote TEXT --vote-n 9`
`report`	`--prompt --no-plots` (writes `results/report.md`)
`chat`	interactive; type a strategy name (e.g. `topp`) to switch live, `quit` to exit
`train`	`--steps 300 --batch-size 24 --lr 3e-3 --block-size 64 --n-embd 128 --n-head 4 --n-layer 2 --seed 123 --corpus --checkpoint`

Appendix B — Metric Formula Cheat Sheet

Name	Definition	Range / direction
distinct-n	|unique n-grams| / |all n-grams|	0..1, higher = more diverse
repetition	1 - distinct-3	0..1, lower = less looping
randomness	mean(step entropy) / ln(vocab)	0 = greedy, 1 = uniform
step entropy	-sum(p * ln p) over the final distribution (nats)	0 = certain, ln(V) = maximum
perplexity	exp(-mean log-probability) under the backend	>= 1, lower = smoother
continuity	output bigrams also found in the corpus / all output bigrams	0..1, higher = more grounded
coherence	0.5·clamp(1 - ln(ppl)/ln 1000) + 0.5·continuity	0..1
relevance	0.5·prompt content-word recall + 0.5·topic keyword fit	0..1
factuality	verified / (verified + unverifiable) - 0.75·contradictions	0..1, clamped
safety	1 - 0.5·(total detector flags)	0..1
helpfulness	0.30·rel + 0.25·coh + 0.20·fact + 0.25·safety	0..1, headline score
candidate score (compute)	0.35·rel + 0.25·coh + 0.20·fact + 0.20·safe	0..1, used by best-of-N

Appendix C — Glossary

Term	Meaning
Attention	Mechanism letting each token weigh how much every earlier token matters to it.
Backend	The component that turns text into logits; interchangeable (ngram/hf/tiny).
Checkpoint	Saved model weights, here `artifacts/tiny_gpt.pt`.
Coherence	How smoothly and plausibly the text flows; blend of fluency and corpus continuity.
Context window	Maximum tokens a model can condition on (tiny: 64; GPT-2: 1024).
Decoding	The rule that picks a token from a probability distribution.
Entropy	Average uncertainty of a distribution; high = the model is undecided.
Factuality	Degree to which claims are supported by the bundled knowledge base.
Hallucination	Confident, fluent output that is unsupported or false.
Logit	Raw (pre-softmax) score for one token.
Nucleus sampling	Top-p: keep the smallest set of tokens covering probability p.
Overfitting	Memorising training data instead of learning patterns.
Perplexity	exp of the average negative log-probability; a surprise score.
Prompt injection	Smuggled instructions inside input or retrieved data that try to hijack behaviour.
Reranker / verifier	A scorer used to pick the best of several sampled candidates.
Self-consistency	Sampling several chains/answers and taking the majority vote.
Softmax	Function converting logits into probabilities that sum to 1.
Temperature	Divides logits before softmax: below 1 sharpens, above 1 flattens.
Test-time compute	Extra computation spent after the prompt (sampling, ranking, thinking longer).
Token	Atomic unit of text the model reads; a word piece with an integer ID.
Transformer	The attention-based architecture behind all modern LLMs.
Vocabulary	The full set of tokens a model can emit.

Appendix D — Dependencies and Environment Notes

Package	Required?	Enables	Installed with
(standard library)	yes	everything: decoding, metrics, safety, compute, n-gram backend, tests, CLI	—
numpy	optional	numeric helpers used by some report code	`pip install -r requirements.txt`
matplotlib	optional	all charts written to `results/*.png`	`pip install -r requirements.txt`
torch	optional	`--backend tiny`, `--backend hf`, the `train` command	`pip install torch`
transformers	optional	`--backend hf` (pretrained GPT-2 and any other causal LM)	`pip install transformers`
reportlab	optional	generating this PDF (`python build_guide_pdf.py`)	`pip install reportlab`

End of guide

You now have the whole picture: the theory (Chapter 2), the map (Chapter 3), the setup (Chapter 4), every algorithm and formula (Chapter 5), the build order (Chapter 6), how to run and read the lab (Chapter 7) and where to go next (Chapter 8). Start with `python run_lab.py compare` and change one number at a time.